JOHNS HOPKINS
WHITING SCHOOL of ENGINEERING

FPGA Design Using VHDL

Module 2E: Conditionals: CSA vs. Process

Process Statements and Conditionals

- Reminders from Lecture 1
  - What are two types of conditionals we covered so far?

```
y <= a when sel = '0' else b;


with data select
        out_disp <=       "1001" when "0000",
                          "0011" when "0001",
                          "1101" when "0010",
                          "1111" when "0011",
                          "1100" when others;
```

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Process Statements and Conditionals Continued

- The following conditional statements are used with **process** statements:

1. **`if/then/else`**
2. **`case/when/is`**

```
show_if : process(a,b,c)
begin
   if(c = b) then
      dataout <= "11111111";
   elsif(a > b) then
      dataout <= a;
   else
      dataout <= b;
   end if;
end process show_if;
```

```
show_case : process(sel,a,b)
begin
   case sel is
      when "00" =>
         nibble_out <= a(3 downto 0);
      when "01" =>
         nibble_out <= a(7 downto 4);
      when "10" =>
         nibble_out <= b(3 downto 0);
      when others =>
         nibble_out <= b(7 downto 4);
   end case;
end process show_case;
```

You don't always need to end with "`else`"

You don't always need to end with "`others`" if all conditionals specified

More on Processes

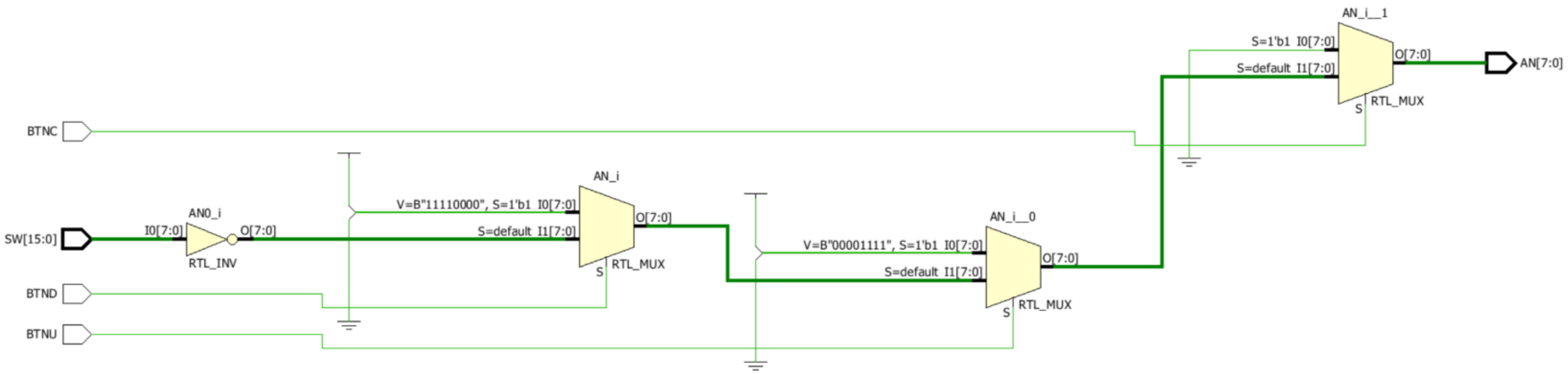
Without Processes:

```
AN <= "00000000" when BTNC = '1' else
      "00001111" when BTNU = '1' else
      "11110000" when BTND = '1' else
      not sw(11 downto 4);
```

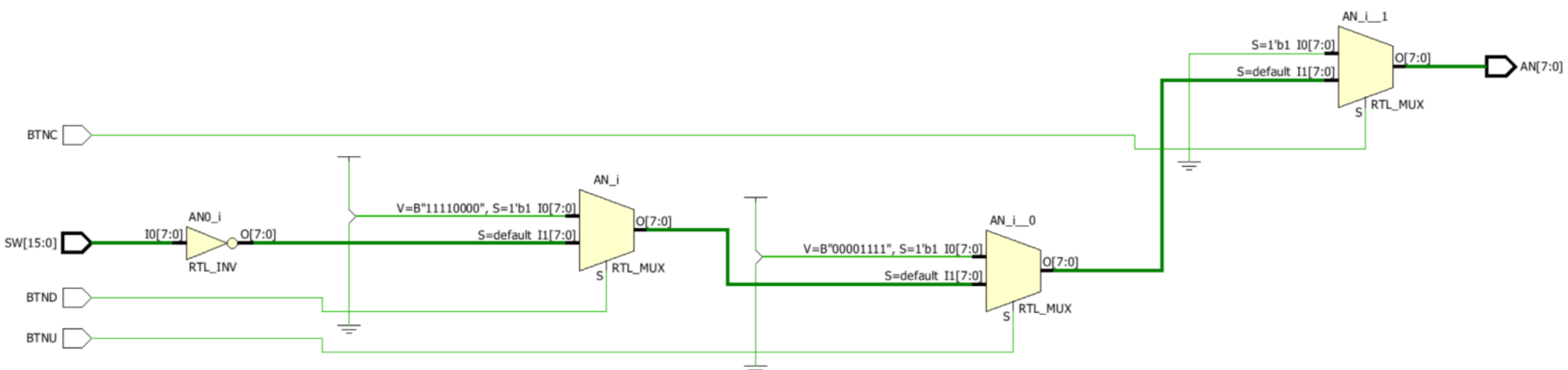
With Processes:

```
process(BTNC, BTNU, BTND, sw)
begin
        if(BTNC = '1') then
            AN <= "00000000";
        elsif(BTNU = '1') then
            AN <= "00001111";
        elsif(BTND = '1') then
            AN <= "11110000";
        else
            AN <= not sw(11 downto 4);
        end if;
end process;
```

And More on Processes



With Processes:



Same Schematic and hardware utilization

Even More on Processes

- Utilization Summaries

Utilization with when/else statement:

Site Type	Used	Fixed	Available	Util%
Slice LUTs*	4	0	63400	<0.01
LUT as Logic	4	0	63400	<0.01
LUT as Memory	0	0	19000	0.00
Slice Registers	0	0	126800	0.00
Register as Flip Flop	0	0	126800	0.00
Register as Latch	0	0	126800	0.00
F7 Muxes	0	0	31700	0.00
F8 Muxes	0	0	15850	0.00

Utilization with if/else process statement:

Site Type	Used	Fixed	Available	Util%
Slice LUTs*	4	0	63400	<0.01
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Conditionals: CSA vs. Process

Without Processes:

```
AN <= "00000000" when BTNC = '1' else
      "00001111" when BTNU = '1' else
      "11110000" when BTND = '1' else
      not sw(11 downto 4);
```

When are they the same?

With Processes:

```
process(BTNC, BTNU, BTND)
begin
        if(BTNC = '1') then
            AN <= "00000000";
        elsif(BTNU = '1') then
            AN <= "00001111";
        elsif(BTND = '1') then
            AN <= "11110000";
        else
            AN <= not sw(11 downto 4);
        end if;
end process;
```

Conditionals: CSA vs. Process Continued

Without Processes:

```
AN <= "00000000" when BTNC = '1' else
      "00001111" when BTNU = '1' else
      "11110000" when BTND = '1' else
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```

With Processes:

```
process(BTNC, BTNU, BTND)
begin
        if(BTNC = '1') then
            AN <= "00000000";
        elsif(BTNU = '1') then
            AN <= "00001111";
        elsif(BTND = '1') then
            AN <= "11110000";
        else
            AN <= not sw(11 downto 4);
        end if;
end process;
```

When are they **NOT** the same?

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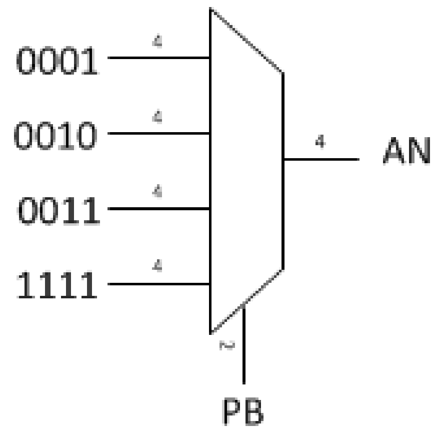
Whiting School of Engineering

Test your knowledge

- Write the VHDL code for a conditional that describes the following table:

PB	AN
00	0001
01	0010
10	0011
11	1111



- Use both types of conditional statements covered in this lecture
    - `if-then-else` and `case-when-is`